Embedded Systems & IoT April -July 2026 Lab Exercise: Schematics with ESP32

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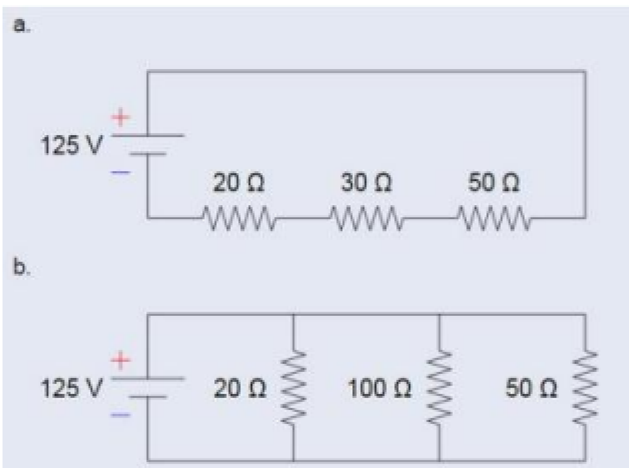
ICS 4E

Part A



For each of these circuits:

1. Calculate the total resistance in ohms,
   - **A. Total Resistance = R1 + R2 +R3**
     20+ 30+ 50 = 100Ω
   - **B. Total Resistance = 1/R1 + 1/R2 + 1/R3**
     1/20 + 1/100 +1/50 =0.08
     1/0.08 =12.5Ω
2. Calculate the total current from the power supply
   - **A. I= V/R**
     I=125/100 =1.25A
   - **B. I=V/R**
     125/12.5 =10A
3. Calculate the current through each resistor
   - **A. I1=I2=I3= Total current= 1.25A**
   - **B. I1 = 125/20 =6.25A**
     I2 = 125/100 =1.25A
     I3 = 125/50 =2.5A

4. Calculate the Voltage drop through each resistor

A. **V=IR**
   V1= 1.25 * 20 =25V
   V2 = 1.25 * 30 =37.5V
   VV3 = 1.25 * 50 =62.5v
B. **V1=V2=V3=Total Voltage= 125V**

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)

A. **P=VI**
   P1 = 25 * 1.25 =31.25W
   P2 = 37.5 * 1.25 = 46.875W
   P3 = 62.5 * 1.25 =78.128W
B. **P=VI**
   P1 = 125 * 6.25 =781.25W
   P2 = 125 * 1.25 = 156.25W
   P3 = 125 * 2.5 =312.5W